

SREENIDHI UNIVERSITY HYDERABAD

(Approved by Telangana State Government G.O.Ms. No.55/2024 and UGC)

Sreenidhi School of Engineering



CURRICULUM STRUCTURE AND SYLLABUS

4 Year B.Tech. CSE Undergraduate Program

A24 Admitted Batch | R26 Regulation

R26 Academic Rules and Regulations

Applicable to students admitted in the AY A24(2024–25), and to lateral entry students admitted in the A25(AY 2025–26), for B.Tech. CSE and CSE AIML Programmes in the Sreenidhi School of Engineering.

<https://drive.google.com/file/d/1zhBKnWl3Y4xn2vwZx45E3JQBuOjx477E/view?usp=sharing>

Note: These R26 Academic Regulations supersede the earlier R24 and R25 Academic Regulations and shall also be applicable to students admitted under the A24 and A25 batches, unless otherwise notified by the University.

SREENIDHI UNIVERSITY

Vision

To be a world class university that produces outstanding professionals and socially responsible graduates who can thrive in a multi-disciplinary environment for the development of the nation and beyond.

Mission

- Foster Excellence through innovative teaching-learning methodologies, rigorous research, and state-of-the-art infrastructure that empowers students to become lifelong learners and thought leaders.
- Advance Knowledge by conducting cutting-edge research that addresses global challenges, creates new knowledge frontiers, and contributes to scientific and technological advancements.
- Catalyze Innovation by nurturing an entrepreneurial ecosystem that transforms ideas into impactful solutions, fosters industry collaborations, and creates value for society.
- Cultivate Global Citizens by providing an inclusive, diverse learning environment that builds cross-cultural competencies, ethical reasoning, and a commitment to sustainable development.
- Serve Society by applying expertise and resources to enhance community well-being, inform policy decisions, and contribute to regional and national development priorities.

Through these commitments, we prepare professionally competent and socially responsible graduates who excel in multidisciplinary environments and contribute meaningfully to national and global advancement

SREENIDHI SCHOOL OF ENGINEERING

Vision

To emerge as a global hub for innovation-driven engineering education and interdisciplinary research, fostering responsible technologists and future leaders

Mission

- To deliver a flexible, industry-aligned curriculum integrating theory with hands-on learning, leveraging cutting-edge infrastructure and pedagogy.
- To foster collaborative research and innovation through strategic partnerships with academia, industry, and global institutions.
- To empower students with leadership, critical thinking, and ethical decision-making skills to tackle real-world challenges.
- To develop socially responsible graduates with global awareness, effective communication skills, and an entrepreneurial mindset for success in diverse careers.

COMPUTER SCIENCE AND ENGINEERING

Vision

To excel in technology and innovation, nurturing competent professionals, pioneering researchers, and socially responsible leaders who contribute to the advancement of the digital world and societal well-being in alignment with global and national aspirations.

Mission

- To foster academic excellence through outcome-based education, innovative teaching methodologies, continuous curriculum enhancement, and integration of emerging technologies to develop industry-ready graduates and lifelong learners.
- To encourage research, innovation, and entrepreneurship by providing state-of-the-art infrastructure and fostering collaborative engagements with industry and academia to address real-world challenges.
- To develop socially responsible and ethically aware graduates with strong communication, leadership, and cross-cultural competencies, capable of contributing meaningfully to a diverse and global workforce.
- To leverage computing and data-driven technologies to address societal needs, promote sustainable development, and contribute to national and international growth.

B.Tech. Computer Science and Engineering

(A24 Admitted Batch | R26 Regulation)

Programme Educational Objectives (PEOs)

PEO #	Programme Educational Objectives
PEO 1	Students will become professionals capable of applying core principles of Computer Science and Engineering to solve real-world problems using emerging technologies.
PEO 2	Students will pursue successful careers as professionals, innovators, or entrepreneurs, including software development, system design, and technology implementation, contributing to industry advancements.
PEO 3	Students will demonstrate ethical practices, social responsibility, and an awareness of the societal impacts of emerging technologies, ensuring their work aligns with global development goals.
PEO 4	Students will engage in interdisciplinary collaborations with peers from diverse fields, enhancing their problem-solving capabilities and contributing to the technological, economic, and social growth of the country.

PEO Articulation Matrix

	PEO1	PEO2	PEO3	PEO4
M1				
M2				
M3				
M4				

H – High, M – Medium, L – Low

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs)

At the end of the Programme the students would be able to:

PO #	Programme Outcome
PO 1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation for the solution of complex engineering problems.
PO 2	Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO 4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 8	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 9	Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 10	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 11	Life-long learning: Recognise the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO #	Programme Specific Outcome - CSE
PSO 1	An ability to design and develop software solutions to address complex engineering problems across various domains using core computer science and engineering principles.
PSO 2	An ability to apply advanced algorithms to optimize system performance and enhance decision-making processes and drive efficiency in a wide range of applications.

Curriculum

Key Features

1. University Core courses are fundamental and mandated by the University to provide a broad and essential foundation across disciplines. Need to earn 17 credits under this category and below are the categories
 - a. Ability Enhancement Courses are designed to improve essential communication, reasoning, and analytical skills, enhancing overall academic and professional competencies.
 - b. Skill Enhancement Courses are practical-focused courses aimed at building technical, vocational, and soft skills relevant to employability and career growth.
 - c. Audit Courses such as Indian Constitution, Sports, UHV-I and II etc. are mandatory non-credit courses undertaken voluntarily for knowledge enhancement without impacting grades or GPA.
 - d. Humanities and Social Sciences courses promote critical thinking, design thinking, innovation and holistic education
2. Faculty Core courses specific to a Basic Science and Engineering faculty aimed at building foundational knowledge in their respective domain. Need to earn 64 credits under this category and below are the categories
 - a. Basic Sciences courses are the foundational science that provide the basic principles and theories essential for advanced learning.
 - b. Engineering Sciences courses are the core engineering courses covering fundamental concepts and applications across various technical disciplines.
 - c. Research, Project and Internship courses are the academic initiatives combining research, project work, and internships to foster practical problem-solving skills specific to B.Tech. CSE.
3. Programme Core courses are tailored to a specific program, focusing on essential knowledge and skills of B.Tech. CSE. Need to earn 57 credits under this category.
 - Need to earn 35 credits under the standard courses.
 - Need to earn 16 credits under their chosen track under programme core
 - Need to earn 6 credits under Skill Development courses which are industry-aligned courses that enhance professional, technical, and employability skills in students
 - The list of tracks under the Programme core that may be offered are:
 - i. Systems & Architectures (SYS)

- ii. Computer Vision (CMV)
- iii. Software Engineering (SWE)
- iv. Artificial Intelligence & Machine Learning (AML)

Note: B.Tech. CSE students must select a track of their interest by the end of Semester 4, with the option to choose General track where they can choose any one course from each track before the end of Semester 5, subject to Department and Dean approval.

4. Programme Electives are the specialized elective courses offering advanced knowledge and skills related to the professional domain.
 - Need to earn 16 credits by opting courses from anyone/multiple segment(s)/general PEs from the offered list of courses by the Dept/School
 - The list of segments under the Programme electives that may be offered are:
 - i. Data Science (DSC)
 - ii. Block Chain (BKC)
 - iii. Cyber Security (CYS)
 - iv. Drone Technologies (DRT)
 - v. General
5. Open Electives are the flexible elective courses chosen by students from a wide range of disciplines to explore diverse interests.
 - Need to earn 9 credits by opting courses from anyone/multiple categories OEs from the offered list of courses by the Dept/School.
 - The list of categories under the Open electives that may be offered are:
 - i. Biotechnology (BTN)
 - ii. Civil Engineering (CVE)
 - iii. Electrical & Electronics Engineering (EEC)
 - iv. Mechanical Engineering (MEC)
 - v. Industrial Automation (IAU)
 - vi. Innovation, Entrepreneurship & Management (IEM)
 - vii. Indian and Foreign Languages (IFL)
 - viii. Science and Technology (SCT)
 - ix. Mathematical Computing (MAC)

Minimum Credit Requirements for the Award of B.Tech. Degree

Category	Abbr.	Subcategory	Abbr.	Courses	Credits	Total Credits
University Core	UC	Ability Enhancement Courses	AEC	4	7	17
		Skill Enhancement Courses	SEC	5	6	
		Audit Courses	AUC	6	0	
		Humanities and Social Sciences	HAS	4	4	
Faculty Core	FC	Basic Sciences	BSC	7	23	64
		Engineering Sciences	ESC	8	22	
		Research, Project and Internship	RPI	5	19	
Programme Core	PC	Standard Courses	SDD	9	35	57
		Track1: Systems & Architectures	SYS	1	4	
		Track 2: Computer Vision	CMV	1	4	
		Track 3: Software Engineering	SWE	1	4	
		Track 4: Artificial Intelligence & Machine Learning	AML	1	4	
		Skill Development	SDC	3	6	
Programme Electives	PE	DSC/BKC/CYS/DRT/General	PE 1	1	4	16
		DSC/BKC/CYS/DRT/General	PE 2	1	4	
		DSC/BKC/CYS/DRT/General	PE 3	1	4	
		DSC/BKC/CYS/DRT/General	PE 4	1	4	
Open Electives	OE	BTN/CVE/MEC/EEC/IAU/IEM/IFL/SCT/MAC	OE 1	1	3	9
		BTN/CVE/MEC/EEC/IAU/IEM/IFL/SCT/MAC	OE 2	1	3	
		BTN/CVE/MEC/EEC/IAU/IEM/IFL/SCT/MAC	OE 3	1	3	
Total Courses and Credits for B.Tech. CSE				62		163

University Core

Need to earn 17 credits from various subcategories of the University Core

University Core (UC)									
Ability Enhancement Courses (AEC)									
Course Code	Level	Courses	L	T	P	S	I	C	CH
24ENG104	100	English for Skill Enhancement	2	0	2	0	0	3	4
24ENGL106	100	Oral Communication Skills Lab - I	0	0	2	0	0	1	2
24ENGL206	100	Oral Communication Lab - II	0	0	2	2	0	2	4
24ENGL605	300	Communication Skills - III	0	0	2	2	0	2	4
Skill Enhancement Courses (SEC)									
24SEC306	100	Soft Skills 1 -Quantitative Aptitude and Personal Effectiveness	0	0	0	2	0	1	2
24SEC406	100	Soft Skills 2 -Reasoning, Verbal Ability and Emotional Intelligence	0	0	0	2	0	1	2
24SEC506	300	Soft Skill 3	0	0	2	2	0	1	2
24SEC307	200	Competitive Programming Practicum	0	0	0	2	0	1	2
24SEC507	300	Code Challenge Accelerator Program	0	0	0	2	0	1	2
Audit Courses (AUC) ^									
24MC30X	100	Sports Bucket	0	0	0	2	0	0	2
	100	Health and Wellness Bucket	0	0	0	2	0	0	2
24MC40X	200	Professional Societies Activity Bucket	0	0	0	2	0	0	2
	200	Community Services Bucket	0	0	0	2	0	0	2
24MC110	100	Induction Program	1	0	0	0	0	0	1
24MC507	100	Indian Constitution	2	0	0	0	0	0	2
24MC704	100	Environmental Studies	2	0	0	0	0	0	2
24MC705	200	Universal Human Values-II	2	0	0	0	0	0	2
Sports Bucket Courses									
24MC301	100	Throwball	0	0	0	2	0	0	2
24MC302	100	Volleyball	0	0	0	2	0	0	2
24MC303	100	Kabaddi	0	0	0	2	0	0	2
24MC304	100	Kho Kho	0	0	0	2	0	0	2
24MC305	100	Table Tennis	0	0	0	2	0	0	2

Health and Wellness Bucket Courses									
24MC306	100	Yoga and Self Defence	0	0	0	2	0	0	2
24MC307	100	Nutrition and Wellbeing	0	0	0	2	0	0	2
Professional Societies Activity Bucket Courses									
24MC401	100	Member	0	0	0	2	0	0	2
24MC402	100	Participant	0	0	0	2	0	0	2
Community Services Bucket Courses									
24MC403	100	Organizer	0	0	0	2	0	0	2
24MC404	100	Volunteer	0	0	0	2	0	0	2
Humanities and Social Sciences (HAS)									
24HASL207	100	Sustainability through Design Thinking	0	0	0	2	0	1	2
24HASL308	100	Marketing for Sustainability	0	0	0	2	0	1	2
24HASL407	200	Innovation and Entrepreneurship - I	0	0	0	2	0	1	2
24HASL508	300	Innovation and Entrepreneurship - II	0	0	0	2	0	1	2

^ Audit Courses are mandatory non-credit courses opted by the students for the award of their BTech program. These courses are evaluated as Satisfactory / Not Satisfactory and they do not contribute to the CGPA, successful completion is essential for fulfilling graduation requirements.

Faculty Core

Need to earn 64 credits from various subcategories of the Faculty Core

Faculty Core (FC)									
Basic Science Courses (BSC)									
Course Code	Level	Courses	L	T	P	S	I	C	CH
24MAT103	100	Calculus	3	0	0	0	0	3	3
24MAT202	100	Discrete Mathematics	3	1	0	0	0	4	4
24MAT301	200	Linear Algebra	3	1	0	0	0	4	4
24MAT401	200	Probability and Statistics	3	1	0	0	0	4	4
24PHY101	100	Engineering Physics	3	0	0	0	0	3	3
24PHYL107	100	Engineering Physics Lab	0	0	4	0	0	2	4
24CHE201	100	Environmental Chemistry & Engineering Biology	3	0	0	0	0	3	3
Engineering Science Courses (ESC)									
24CSE102	100	Problem Solving and Computer Programming through C++	3	0	0	0	0	3	3

24CSEL108	100	Problem Solving and Computer Programming through C++ Lab	0	0	4	0	0	2	4
24CSEL205	100	Programming through Python	0	0	4	2	0	3	6
24CSEJ204	100	Data Structures through C++	2	0	4	0	0	4	6
24EEE105	100	Elements of Electrical and Electronics Engineering	3	0	2	0	0	4	5
24CSE203	100	Computer Organization and Architecture	3	0	0	0	0	3	3
24CSEL207	100	IT Workshop	0	0	2	0	0	1	2
24ECE405	200	Microprocessors & Microcontrollers	3	0	0	0	0	3	3
Research, Project and Internship Courses (RPI)									
24CSEI222	100	Social Internship	0	0	0	0	3	1	3
24CSE444	200	Technical Writing	0	0	0	0	3	1	3
24CSEI666	200	Summer Internship	0	0	0	0	4	1	4
24CSEI777	300	Capstone Project - 1/Internship-1	0	0	0	0	4	5	4
24CSEI888	400	Capstone Project - 2/Internship-2/ Research	0	0	0	0	8	10	8

Programme Core

Need to earn 57 credits from various subcategories of the Programme Core

Programme Core (PC)									
Standard Courses (SDD)									
Course Code	Level	Courses	L	T	P	S	I	C	CH
24CSEL109	100	Fundamentals of Web Technologies	0	0	4	0	0	2	4
24CSEJ303	200	Object Oriented Programming through Java	2	0	4	0	0	4	6
24CSEJ304	200	Operating Systems	3	0	2	0	0	4	5
24CSEJ305	200	Design and Analysis of Algorithms	3	0	2	0	0	4	5
24CSEJ302	200	Database Management Systems	3	0	2	0	0	4	5
24CSEJ402	200	Automata and Compiler Design	3	0	2	0	0	4	5
24AMLJ404	200	Artificial Intelligence & Machine Learning	4	0	2	0	0	5	6
24CSEJ601	200	Software Engineering	3	0	2	0	0	4	5
24CSEJ501	300	Computer Networks	3	1	0	0	0	4	4
Skill Development Courses (SDC)									
24CSES405	200	Full Stack Application Development	0	0	2	2	0	2	4

24CSES507	300	Coding Excellence Program I	0	0	2	2	0	2	4
24CSES606	300	Coding Excellence Program II	0	0	2	2	0	2	4
Track 1: Systems & Architectures (SYS)									
24ECEJ502	200	Embedded Systems	3	0	2	0	0	4	5
24CSEJ512	300	Distributed Systems	3	0	2	0	0	4	5
24CSEJ522	300	Advanced Computer Architecture	3	0	2	0	0	4	5
24AMLJ532	400	GPU Programming and Architectures	3	0	2	0	0	4	5
Track 2: Computer Vision (CMV)									
24CSEJ503	200	Human Computer Interaction	3	0	2	0	0	4	5
24CSEJ513	200	Computer Graphics and Multimedia	3	0	2	0	0	4	5
24CSEJ523	300	Image Processing	3	0	2	0	0	4	5
24AMLJ502	300	Computer Vision through Open CV	3	0	2	0	0	4	5
Track 3: Software Engineering (SWE)									
24CSEJ602	200	Cloud Computing	3	0	2	0	0	4	5
24CSEJ612	200	Web Application Frameworks	3	0	2	0	0	4	5
24CSEJ622	300	Middleware Technologies	3	0	2	0	0	4	5
24CSEJ632	300	Principles of MLOps	3	0	2	0	0	4	5
Track 4: Artificial Intelligence & Machine Learning (AML)									
24AMLJ502	300	Computer Vision through Open CV	3	0	2	0	0	4	5
24AMLJ602	300	Deep Learning	3	0	2	0	0	4	5
24AMLJ503	300	Natural Language Processing	3	0	2	0	0	4	5
24AMLJ701	400	Generative AI	3	0	2	0	0	4	5

Programme Electives

Need to earn 16 credits by opting courses from anyone/multiple segment(s)/general PEs from the offered list of courses by the Dept/School

Programme Electives (PE)									
Segment 1: Data Science (DSC)									
24DSCJ504	200	Data Visualization with R	3	0	0	2	0	4	5
24DSCJ603	300	Applied Data Science	3	0	0	2	0	4	5
24DSCJ604	300	Health Informatics	3	0	0	2	0	4	5
24DSCJ702	400	Business Analytics	3	0	0	2	0	4	5

Segment 2: Block Chain (BKC)									
24BKSJ504	200	Smart Contracts	3	0	0	2	0	4	5
24BKCJ603	200	Blockchain Security	3	0	0	2	0	4	5
24BKCJ604	300	Blockchain Business Model	3	0	0	2	0	4	5
24BKCJ702	400	Advanced Blockchain Architectures	3	0	0	2	0	4	5
Segment 3: Cyber Security (CYS)									
24CYSJ504	300	<u>Cryptography and Network Security</u>	3	0	0	2	0	4	5
24CYSJ603	300	Foundations of Cyber Security	3	0	0	2	0	4	5
24CYSJ604	300	Threat Intelligence	3	0	0	2	0	4	5
24CYSJ702	300	Cyber Forensics	3	0	0	2	0	4	5
Segment 4: Drone Technologies (DRT)									
24ECEJ504	200	Drone Design and Manufacturing	3	0	0	2	0	4	5
24ECEJ603	300	Drone Navigation and Control	3	0	0	2	0	4	5
24ECEJ604	300	Legal and Ethical Aspects of Drone Technology	3	0	0	2	0	4	5
24ECEJ702	400	Drone Applications: Agriculture & Disaster Management	3	0	0	2	0	4	5
General Track									
24CSEP504	300	Foundations of Metaverse	3	0	0	2	0	4	5
24CSEP514	300	Foundations of UX Design	3	0	0	2	0	4	5
24AML504	300	Data Warehousing and Data Mining	3	0	0	2	0	4	5
24CSEP524	300	Ad hoc and Sensor Networks	3	0	0	2	0	4	5
24CSEP603	300	IoT Architecture and Protocols	3	0	0	2	0	4	5
24CSEP613	300	Information Security and Risk Management	3	0	0	2	0	4	5
24CSEP623	300	Advanced Data Structures	3	0	0	2	0	4	5
24CSEP633	300	Software Testing Methodologies	3	0	0	2	0	4	5
24CSEP604	300	Design Patterns	3	0	0	2	0	4	5
24CYSP614	300	Introduction to Cyber Physical Systems	3	0	0	2	0	4	5
24CSEP624	300	Real Time Systems	3	0	0	2	0	4	5
24CSEP634	300	Robotics and Automation	3	0	0	2	0	4	5
24CSEP702	400	Agile Software Development	3	0	0	2	0	4	5
24CSEP712	400	Quantum Computing	3	0	0	2	0	4	5
24CSEP722	400	Operating System Security	3	0	0	2	0	4	5
24CSEP732	400	Intrusion Detection and Prevention Systems	3	0	0	2	0	4	5

24CSEP742	400	Genomic Data Science and Clustering	3	0	0	2	0	4	5
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Open Electives (OE)

Need to earn 9 credits by opting courses offered by various Depts.

Open Electives (OE)									
Biotechnology (BTN)									
Course Code	Level	Courses	L	T	P	S	I	C	CH
24BTNO505	200	Biotechnology for Society	3	0	0	0	0	3	3
24BTNO703	200	Nano Biotechnology	3	0	0	0	0	3	3
24BTNO704	200	Fundamentals of Computational Biology	3	0	0	0	0	3	3
Civil Engineering (CVE)									
24CVLO505	100	Fundamentals of Remote Sensing	3	0	0	0	0	3	3
24CVLO703	100	Fundamentals of GIS	3	0	0	0	0	3	3
24CVLO704	100	Disaster Management	3	0	0	0	0	3	3
Electrical & Electronics Engineering (EEC)									
24EEEO505	100	Arduino for Beginners	3	0	0	0	0	3	3
24EEEO703	200	Fundamentals of Autonomous Vehicles	3	0	0	0	0	3	3
24EEEO704	300	Introduction to Wireless Networks	3	0	0	0	0	3	3
Mechanical Engineering (MEC)									
24MECO505	200	Logistics and Supply Chain Management	3	0	0	0	0	3	3
24MECO703	300	Energy Conservation and Management	3	0	0	0	0	3	3
24MECO704	200	Fundamentals of Project Management	3	0	0	0	0	3	3
Industrial Automation (IAU)									
24IAUO505	300	Industry 4.0	3	0	0	0	0	3	3
24IAUO703	300	Control Engineering	3	0	0	0	0	3	3
24IAUO704	300	Smart Manufacturing	3	0	0	0	0	3	3
Innovation, Entrepreneurship & Management (IEM)									
24HASL505	300	Startup Launch- I	3	0	0	0	0	3	3
24HASL705	300	Startup Launch - II	3	0	0	0	0	3	3
24IEMO505	100	Marketing Management	3	0	0	0	0	3	3
24IEMO703	200	Organizational Behaviour	3	0	0	0	0	3	3

24IEMO801	300	Sales and Distribution Management	3	0	0	0	0	3	3
Indian and Foreign Languages (IFL)									
24IFLO505	200	Sanskrit	3	0	0	0	0	3	3
24IFLO506	200	Telugu	3	0	0	0	0	3	3
24IFLO703	200	German	3	0	0	0	0	3	3
24IFLO704	200	Spanish	3	0	0	0	0	3	3
Science and Technology (SCT)									
24SCTO505	300	Introduction to Quantum Communication	3	0	0	0	0	3	3
24SCTO703	300	Introduction to Quantum Sensing	3	0	0	0	0	3	3
24SCTO704	300	Introduction to Quantum Materials and Devices	3	0	0	0	0	3	3
Mathematical Computing (MAC)									
24MACO505	300	Operations Research	3	0	0	0	0	3	3
24MACO703	300	Numerical Methods	3	0	0	0	0	3	3
24MACO704	300	Differential Equations and Transform Techniques	3	0	0	0	0	3	3

Programme Structure

A24 B.Tech. CSE Structure - Suggested**Semester 1**

Cat	Course Code	Courses	L	T	P	S	I	C	CH
FC	24PHY101	Engineering Physics ^b	3	0	0	0	0	3	3
FC	24CHE201	Environmental Chemistry & Engineering Biology ^b							
FC	24CSE102	Problem Solving and Computer Programming through C++	3	0	0	0	0	3	3
FC	24MAT103	Calculus	2	1	0	0	0	3	3
UC	24ENG104	English for Skill Enhancement	2	0	0	0	0	2	2
FC	24EEE105	Elements of Electrical and Electronics Engineering	3	0	0	0	0	3	3
UC	24ENGL106	Oral Communication Skills Lab - I	0	0	2	0	0	1	2
FC	24PHYL107	Engineering Physics Lab [Engg Physics Comb] ^b	0	0	4	0	0	2	4
PC	24CSEL109	Fundamentals of Web Technologies [Env Chem comb] ^b							
FC	24CSEL108	Problem Solving and Computer Programming through C++ Lab	0	0	4	0	0	2	4
FC	24CSEL207	IT Workshop	0	0	2	0	0	1	2
UC	24MC110	Induction Program ^a	1	0	0	0	0	0	1
Total for Semester 1			14	1	12	0	0	20	27

Semester 2

Cat	Course Code	Courses	L	T	P	S	I	C	CH
FC	24PHY101	Engineering Physics ^b	3	0	0	0	0	3	3
FC	24CHE201	Environmental Chemistry & Engineering Biology ^b							
FC	24MAT202	Discrete Mathematics	3	1	0	0	0	4	4
FC	24CSE203	Computer Organization and Architecture	3	0	0	0	0	3	3
FC	24CSEJ204	Data Structures through C++	2	0	4	0	0	4	6
FC	24CSEL205	Programming through Python	0	0	4	2	0	3	6
UC	24ENGL206	Oral Communication Lab - II	0	0	2	2	0	2	4
FC	24PHYL107	Engineering Physics Lab [Engg Physics Comb] ^b	0	0	4	0	0	2	4
PC	24CSEL109	Fundamentals of Web Technologies [Env Chem comb] ^b							
UC	24HASL207	Sustainability through Design Thinking	0	0	0	2	0	1	2
Total for Semester 2			11	1	14	6	0	22	32
FC	24CSEI222	Social Internship ^d	0	0	0	0	3	1	3

Semester 3

Cat	Course Code	Courses	L	T	P	S	I	C	CH
FC	24MAT301	Linear Algebra	3	1	0	0	0	4	4
PC	24CSEJ302	Database Management Systems	3	0	2	0	0	4	5
PC	24CSEJ303	Object Oriented Programming through Java	2	0	4	0	0	4	6
PC	24CSEJ304	Operating Systems	3	0	2	0	0	4	5
PC	24CSEJ305	Design and Analysis of Algorithms	3	0	2	0	0	4	5
UC	24SEC306	Soft Skills 1 -Quantitative Aptitude and Personal Effectiveness	0	0	0	2	0	1	2
UC	24SEC307	Competitive Programming Practicum	0	0	0	2	0	1	2
UC	Refer UCs	Sports Bucket/Health and Wellness Bucket ^c	0	0	0	2	0	0	2
		Professional Societies /Community Services Bucket ^c							
UC	24HASL308	Marketing for Sustainability	0	0	0	2	0	1	2
Total for Semester 3			14	1	10	8	0	24	33
Semester 4									
Cat	Course Code	Courses	L	T	P	S	I	C	CH
FC	24MAT401	Probability and Statistics	3	1	0	0	0	4	4
PC	24CSEJ402	Automata and Compiler Design	3	0	2	0	0	4	5
PC	24AMLJ404	Artificial Intelligence & Machine Learning	4	0	2	0	0	5	6
FC	24ECE405	Microprocessors & Microcontrollers	3	0	0	0	0	3	3
PC	24CSES405	Full Stack Application Development	0	0	2	2	0	2	4
UC	24SEC406	Soft Skills 2 -Reasoning, Verbal Ability and Emotional Intelligence	0	0	0	2	0	1	2
UC	Refer UCs	Sports Bucket/Health and Wellness Bucket ^c	0	0	0	2	0	0	2
		Professional Societies /Community Services Bucket ^c							
UC	24HASL407	Innovation and Entrepreneurship - I	0	0	0	2	0	1	2
UC	24SEC507	Code Challenge Accelerator Program	0	0	0	2	0	1	2
Total for Semester 4			13	1	6	10	0	21	30
FC	24CSEI444	Technical Writing ^e	0	0	0	0	3	1	
Semester 5									
Cat	Course Code	Courses	L	T	P	S	I	C	CH
PC	24CSEJ601	Software Engineering	3	0	2	0	0	4	5
PC	Refer PCs	Track1/Track2/Track3/Track4/General	3	0	2	0	0	4	5
PC	Refer PCs	Track1/Track2/Track3/Track4/General	3	0	2	0	0	4	5
PE	Refer PEs	Program Elective 1*	3	0	0	2	0	4	5

OE	Refer OEs	Open Elective 1#	3	0	0	0	0	3	3
UC	24SEC506	Soft Skills 3	0	0	2	2	0	2	4
PC	24CSES507	Coding Excellence Program I	0	0	2	2	0	2	4
UC	24HASL508	Innovation and Entrepreneurship - II	0	0	0	2	0	1	2
Total for Semester 5			15	0	10	8	0	25	33
Semester 6									
Cat	Course Code	Courses	L	T	P	S	I	C	CH
PC	24CSEJ501	Computer Networks	3	1	0	0	0	4	4
PC	Refer PCs	Track1/Track2/Track3/Track4/General	3	0	2	0	0	4	5
PE	Refer PEs	Program Elective 2*	3	0	0	2	0	4	5
PE	Refer PEs	Program Elective 3*	3	0	0	2	0	4	5
UC	24ENGL605	Communication Skills - III	0	0	2	2	0	2	4
PC	24CSES606	Coding Excellence Program II	0	0	2	2	0	2	4
UC	24MC607	Indian Constitution	2	0	0	0	0	0	2
Total for Semester 6			14	1	6	8	0	20	29
FC	24CSEI666	Summer Internship ^f	0	0	0	0	4	2	4
Semester 7									
Cat	Course Code	Courses	L	T	P	S	I	C	CH
PC	Refer PCs	Track1/Track2/Track3/Track4/General	3	0	2	0	0	4	5
PE	Refer PEs	Program Elective 4*	3	0	0	2	0	4	5
OE	Refer OEs	Open Elective 2#	3	0	0	0	0	3	3
OE	Refer OEs	Open Elective 3#	3	0	0	0	0	3	3
UC	24MC704	Environmental Studies	2	0	0	0	0	0	2
UC	24MC705	Universal Human Values - II	2	0	0	0	0	0	2
FC	24CSEI777	Capstone Project - 1/Internship-1	0	0	0	0	5	5	5
Total for Semester 7			16	0	2	2	5	21	25
Semester 8									
Cat	Course Code	Courses	L	T	P	S	I	C	CH
FC	24CSEI888	Capstone Project - 2/Internship-2/Research	0	0	0	0	10	10	10
Total for Semester 8			0	0	0	0	10	10	10
Total for B.Tech. CSE			97	5	60	42	25	163	219

Note:

- a. 24MC110 Induction Program conducted during the first two weeks from the commencement of Semester 1.
- b. 24PHY101 Engineering Physics + 24PHY107 Engineering Physics Laboratory and 24CHE101 Environmental Chemistry & Engineering Biology + 24CSEL209 Fundamentals of Web Technologies is offered in both Semester 1 and Semester 2 allowing students the flexibility to complete these courses in either semester
- c. The 24MC30X Sports/Health & Wellness and 24MC40X Professional Societies/Community Services buckets are offered in both Semester 3 and Semester 4, allowing students the flexibility to complete these courses in either semester.
- d. Social Internship to be completed during the vacation after the 1st year and before the commencement of 2nd year and registered in the 3rd semester.
- e. Technical Writing to be completed during the vacation after the 2nd year and before the commencement of 3rd year and registered in the 5th semester.
- f. Summer Internship to be completed during the vacation after the 3rd year and before the commencement of 4th year and registered in the 7th semester.
- g. Program Elective * - May opt from anyone/multiple segment(s)/general PEs from the offered list of courses by the Dept/School.
- h. Open Elective # - May opt from anyone/multiple categories OEs from the offered list of courses by the Dept/School.

(Updated on 03 July 2026)



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